

B13.9 Flit packet definitions

This section defines the flit format. See:

- [B13.9.1 Request flit](#)
- [B13.9.2 Response flit](#)
- [B13.9.3 Snoop flit](#)
- [B13.9.4 Data flit](#)

B13.9.1 Request flit

[Table B13.6](#) shows the Request flit format in a REQ channel packet starting at bit 0.

The following key is used:

MBZ Must Be Zero

Table B13.6: Request flit format

REQFLIT[(R-1):0] format		
Field	Field width	Comments
QoS	4	-
TgtID	7 to 16	Width determined by NodeID_Width
SrcID	7 to 16	Width determined by NodeID_Width
TxnID	12	-
ReturnNID	7 to 16	Used for DMT
StashNID		Used in Stash transactions
{(NodeID_Width - 7)'b0,		MBZ
DataTarget[6:0]}		Used to aid data placement in system cache hierarchy
StashNIDValid	1	Used in Stash transactions
Endian		Used in Atomic transactions
Deep		Used in CleanSharedPersist* transactions
PrefetchTgtHint		Hint for Chip-to-Chip receivers
ReturnTxnID[11:0]	12	Used for DMT
{6'b0,		MBZ
StashLPIDValid,		Used in Stash transactions
StashLPID[4:0]}		Used in Stash transactions
Opcode	7	-
MultiReq	1	Used in multi-request transactions.
NumReq	6	Field determined by the MultiReq value
{3'b0, Size}		
Addr	RAW = 44 to 52	Width determined by Req_Addr_Width (RAW)

Continued on next page

Table B13.6 – Continued from previous page

REQFLIT[(R-1):0] format		
Field	Field width	Comments
PAS	3	-
LikelyShared	1	-
AllowRetry	1	-
Order	2	-
PCrdType	4	-
MemAttr	4	-
SnpAttr	1	-
DoDWT		Used for DWT
PGroupID[7:0]	8	Used in PCMO transactions
StashGroupID[7:0]		Used in the StashOnceSep transaction
TagGroupID[7:0]		Used for Memory Tagging
{3'b0, LPID[4:0]}		MBZ
Excl	1	Used in Exclusive transactions
SnoopMe		Used in Atomic transactions
CAH		Used in CopyBack Write transactions
ExpCompAck	1	-
TagOp	2	-
TraceTag	1	-
MPAM	M = 0	No MPAM bus
	M = 12, 15	-
PBHA	PB = 0	No PBHA bus
	PB = 4	-
MECID	E = 0	No MECID bus
	E = 16	Used for RME-MEC
StreamID	R = 0	No StreamID bus
	R = 16	Used for RME-CDA
		Common field is Max(E,R)
SecSID1	S = 0	No SecSID1 bit
	S = 1	Used for RME-CDA
RSVDC	Y = 0	No RSVDC bus
	Y = 4, 8, 12, 16, 24, 32	Width determined by Req_RSVDC_Width .

Continued on next page

Table B13.6 – Continued from previous page

REQFLIT[(R-1):0] format		
Field	Field width	Comments
Total	(93 + RAW + Y + M + PB + S + Max(E,R)) to (120 + RAW + Y + M + PB + S + Max(E,R))	

B13.9.2 Response flit

Table B13.7 shows the Response flit format in a RSP channel packet starting at bit 0.

The following key is used:

MBZ Must Be Zero

Table B13.7: Response flit format

RSPFLIT[(T-1):0] format		
Field	Field width	Comments
QoS	4	-
TgtID	7 to 16	Width determined by NodeID_Width
SrcID	7 to 16	Width determined by NodeID_Width
TxnID	12	-
Opcode	5	-
RespErr	2	-
Resp	3	-
FwdState[2:0]	3	Used for DCT
{2'b0, DataPull }		MBZ Used in Stash transactions
CBusy	3	-
DBID[11:0]	12	-
{4'b0, PGroupID [7:0]}		MBZ Used in Persistent CMO transactions
{4'b0, StashGroupID [7:0]}		MBZ Used in Stash transactions
{4'b0, TagGroupID [7:0]}		MBZ Used for Memory Tagging
PCrdType	4	-

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Table B13.7 – Continued from previous page

RSPFLIT[(T-1):0] format		
Field	Field width	Comments
TagOp	2	-
TraceTag	1	-
CacheLineID	6	Used in multi-request transactions
Total	T = 71 to 89	

B13.9.3 Snoop flit

Table B13.8 shows the Snoop flit format in a SNP channel packet starting at bit 0.

The following key is used:

MBZ Must Be Zero

Table B13.8: Snoop flit format

SNPFLIT[(S-1):0] format		
Field	Field width	Comments
QoS	4	-
SrcID	7 to 16	Width determined by NodeID_Width
TxnID	12	-
FwdNID	7 to 16	Width determined by NodeID_Width
{(NodeID_Width - 4)'0, PBHA[3:0]}		
FwdTxnID[11:0]	12	Used for DCT
{6'b0, StashLPIDValid, StashLPID[4:0]}		
{4'b0, VMIDExt[7:0]}		
Opcode	5	-
Addr	SAW = 41 to 49	Req_Addr_Width - 3
PAS	3	-
DoNotGoToSD	1	-
RefToSrc	1	-
TraceTag	1	-
MPAM	M = 0	No MPAM bus

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Table B13.8 – Continued from previous page

SNPFLIT[(S-1):0] format		
Field	Field width	Comments
	M = 12, 15	-
MECID	E = 0	No MECID bus
	E = 16	Used for RME-MEC
Total	S = (53 + SAW + M + E) to (71 + SAW + M + E)	

B13.9.4 Data flit

Table B13.9 shows the Data flit format in a DAT channel packet starting at bit 0.

The number of data flits required is dependent on the number of data bytes, and the data bus width. See B2.9.4 [Data packetization](#).

The data channel interface supports a 128-bit, 256-bit, and 512-bit data bus width. There are three data flit formats defined, one for each of the three data bus widths supported at the data channel interface.

DataCheck field width is either 0 or equal to the width of the Data field divided by 8.

Poison field width is either 0 or equal to the width of the Data field divided by 64.

The following key is used:

MBZ Must Be Zero

Table B13.9: Data flit format

DATFLIT[(D-1):0] format		
Field	Field width	Comments
QoS	4	-
TgtID	7 to 16	Width determined by NodeID_Width
SrcID	7 to 16	Width determined by NodeID_Width
TxnID	12	-
HomeNID	7 to 16	Width determined by NodeID_Width
{(NodeID_Width - 5)'b0, MismatchedMECID, PBHA[3:0]}		
Opcode	4	-
RespErr	2	-
Resp	3	-
DataSource[7:0]	8	Indicates Data source in a response
{5'b0,		MBZ

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Table B13.9 – Continued from previous page

DATFLIT[(D-1):0] format		
Field	Field width	Comments
FwdState[2:0]		Used for DCT
DataPull	1	Used in Stash transactions
CBusy	3	-
MECID	E = 0	No MECID bus
	E = 16	Used for RME-MEC
{4'b0, DBID[11:0]}	16	-
CCID	2	-
DataID	2	-
CacheLineID	6	-
TagOp	2	-
Tag	DW/32 = 4, 8, 16	-
TU	DW/128 = 1, 2, 4	-
TraceTag	1	-
CAH	1	-
NumDat	2	-
Replicate	1	-
RSVDC	Y = 0	No RSVDC bus
	Y = 4, 8, 12, 16, 24, 32	Width determined by Dat_RSVDC_Width .
BE	DW/8 = 16, 32, 64	-
Data	DW = 128, 256, 512	DW = B16.1.13 Data_Width
DataCheck	0 or DW/8 = 16, 32, 64	DC = DataCheck
Poison	0 or DW/64 = 2, 4, 8	P = Poison
Total	D = (240 to 267) + Y + DC + P	DW = 128-bit Data
	D = (389 to 416) + Y + DC + P	DW = 256-bit Data
	D = (687 to 714) + Y + DC + P	DW = 512-bit Data